AIAC
LAB ASSIGNMENT-18.1

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Batch:02

Task 1 – Weather API Integration

Connect to a Weather API to fetch current weather details for a given city.

Instructions:

- Prompt AI to generate Python code using the requests library.
- Handle errors: invalid city name, API key errors, and network failures.
- Display temperature, humidity, and weather description in a readable format.

Expected Output:

- A working Python script that fetches and displays weather data with proper error handling.

Code :

week 18 > demo.py > ...

```python
"""
Weather Fetcher
use open meteo API and  gecoding API
"""
import requests
def get_coordinates(location):
    """Get the coordinates of a location using the geocoding API"""
    url = f"https://geocoding-api.open-meteo.com/v1/search?name={location}"
    response = requests.get(url)
    data = response.json()
    if "results" in data and len(data["results"]) > 0:
        return data["results"][0]["latitude"], data["results"][0]["longitude"]
    else:
        raise ValueError("Location not found")
def get_weather(latitude, longitude):
    """Get the current weather for the given coordinates using the Open Meteo API"""
    url = f"https://api.open-meteo.com/v1/forecast?latitude={latitude}&longitude={longitude}&current_weather=true"
    response = requests.get(url)
    data = response.json()
    if "current_weather" in data:
        return data["current_weather"]
    else:
        raise ValueError("Weather data not found")
def main():
    location = input("Enter a location: ")
    try:
        latitude, longitude = get_coordinates(location)
        weather = get_weather(latitude, longitude)
        print(f"Current weather in {location}:")
        print(f"Temperature: {weather['temperature']}°C")
        print(f"Wind Speed: {weather['windspeed']} km/h")
        print(f"Weather Code: {weather['weathercode']}")
```

Output :

```
PS C:\Users\User\OneDrive\Desktop
IAC/week 18/demo.py"
Enter a location: warangal
Current weather in warangal:
Temperature: 33.3°C
Wind Speed: 1.3 km/h
Weather Code: 2
```

Task 2 – Currency Exchange Rates

Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.

Instructions:

- Use AI to generate API request code.
- Validate user input for amount and currency codes.
- Implement error handling for invalid responses and API downtime.
- Show conversion results clearly in tabular format.

Expected Output:

- A script that converts INR to multiple currencies with graceful error handling.

Code:

week 18 > currency.py > ...

```python
"""
design a currency exchanger that can convert inr to different currencies and vice versa.
use currency exchangr api
"""
import requests
def get_exchange_rate(from_currency, to_currency):
    """Get the exchange rate from one currency to another using the Currency Exchange API"""
    url = f"https://api.exchangerate-api.com/v4/latest/{from_currency}"
    response = requests.get(url)
    data = response.json()
    if "rates" in data and to_currency in data["rates"]:
        return data["rates"][to_currency]
    else:
        raise ValueError("Currency not found")
def convert_currency(amount, from_currency, to_currency):
    """Convert an amount from one currency to another using the exchange rate"""
    exchange_rate = get_exchange_rate(from_currency, to_currency)
    return amount * exchange_rate
```

Output:

```
PS C:\Users\User\OneDrive\Desktop\AIAC> & C:/Users/Use
IAC/week 18/currency.py"
Enter the amount to convert: 100
Enter the currency to convert from (e.g., INR): INR
Enter the currency to convert to (e.g., USD): USD
100.0 INR is equal to 1.06 USD
```

Task 3 – GitHub Repository Info Fetcher

Connect to the GitHub API to fetch details of a repository (e.g., stars, forks, issues).

Instructions:

- Prompt AI to write a Python function to send GET requests to GitHub API.
- Handle rate limit errors, 404 (repo not found), and invalid input.
- Display repository name, description, stars, forks, and open issues.

Expected Output:

- A Python program that retrieves and displays GitHub repository information with robust error handling.

week 18 > github.py > main

```python
write a Python function to send GET requests to GitHub API
"""
import requests
def get_repo_info(owner, repo):
    """Get the details of a GitHub repository using the GitHub API"""
    url = f"https://api.github.com/repos/{owner}/{repo}"
    response = requests.get(url)
    if response.status_code == 200:
        data = response.json()
        return {
            "name": data["name"],
            "owner": data["owner"]["login"],
            "stars": data["stargazers_count"],
            "forks": data["forks_count"],
            "open_issues": data["open_issues_count"]
        }
    else:
        raise ValueError("Repository not found")
```

Output:

```
PS C:\Users\User\OneDrive\Desktop\AIAC> & C:/Users/User/AppData/Local/
IAC/week 18/github.py"
Enter the repository owner: sayyedataj
Enter the repository name: HUNGRY-GO-ONE-CLICK-MEAL-ORDERING-SYSTEM
Repository: HUNGRY-GO-ONE-CLICK-MEAL-ORDERING-SYSTEM
Owner: sayyedataj
Stars: 0
Forks: 0
Open Issues: 0
PS C:\Users\User\OneDrive\Desktop\AIAC>
```

Task 4 – Real-Time Application: News Headlines Aggregator

Scenario: Build a News Aggregator using a free News API.

Requirements:

- Fetch top 5 headlines for a given category (sports, technology, health).
- Handle errors like invalid category, missing API key, and request failures.
- Display headlines in a user-friendly numbered list format.
- Include a retry mechanism if the first request fails.

Expected Output:

- A working Python script that retrieves and displays news headlines with proper error handling

Code:

week 18 > news.py > ...

```python
"""
News Headlines Fetcher
use newsapi.org
"""
import requests
def get_news_headlines(category, api_key):
    """Fetch top 5 news headlines for a given category using the NewsAPI"""
    url = f"https://newsapi.org/v2/top-headlines?category={category}&apiKey={api_key}"
    response = requests.get(url)
    if response.status_code == 200:
        data = response.json()
        if "articles" in data:
            return [article["title"] for article in data["articles"][:5]]
        else:
            raise ValueError("No articles found")
    elif response.status_code == 401:
        raise ValueError("Invalid API key")
    elif response.status_code == 400:
        raise ValueError("Invalid category")
```

Output:

```
PS C:\Users\User\OneDrive\Desktop\AIAC> & C:/Users/User/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/User/One
IAC/week 18/news.py"
Enter your NewsAPI key: 71b8ebd4f2f2456191db37f7faf6413a
Enter the news category (sports, technology, health): sports
Top 5 headlines in sports:
1. 2026 MLB predictions: Playoffs, World Series, MVPs and more - ESPN
2. Miami Open: Coco Gauff reaches semi-finals for first time - BBC
3. Cavs coach Kenny Atkinson sends pointed message to team about playoff fate following win over Orlando - Cleveland.com
4. Team USA's choice for Olympic flag football is obvious - theScore.com
5. Bryan Hodgson's wife stole the March Madness spotlight with shocking comment from new Providence coach - New York Post
```